RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Thursday, 8 October 2026, 4:24 PM
State	Finished
Completed on	Thursday, 8 October 2026, 4:26 PM
Time taken	1 min 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main()
4  {
```

```c
    int n;
    scanf("%d", &n);

    int count = 0;
    int rev = 0, remainder;

    count++;                                /* rev = 0 initialization */

    while (n != 0)
    {
        count++;                            /* loop condition (true) */
        remainder = n % 10;
        count++;                            /* remainder = n % 10 */
        rev = rev * 10 + remainder;
        count++;                            /* rev = rev * 10 + remainder */
        n /= 10;
        count++;                            /* n /= 10 */
    }
    count++;                                /* loop condition (false) */

    count++;                                /* print(rev) */

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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